

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	XXXXX
Project Title	Credit Card Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution is an AI and Machine Learning-based Credit Card Approval Prediction System developed using Python, Flask, HTML, CSS, and Scikit-learn. The system collects applicant details through a web interface, processes the data, and uses a Random Forest Classifier to predict whether the credit card application is likely to be approved or rejected. The trained model is saved using Pickle and loaded into the Flask application to provide instant predictions.

Project Overview

Objective	The primary objective is to develop an AI and Machine Learning based Credit Card Approval Prediction System by implementing advanced machine learning techniques to provide faster and more accurate credit card approval predictions.
Scope	The project analyzes applicant information and predicts credit card approval using a Random Forest machine learning model integrated with a Flask web application, improving the efficiency and accuracy of the approval process.

Problem Statement

Description	The existing credit card approval process is time-consuming and involves manual verification, leading to delays, inconsistent decisions, and customer uncertainty.
Impact	The proposed system improves operational efficiency, reduces manual effort, enables faster and more consistent approval decisions, and enhances customer satisfaction.

Proposed Solution

Approach	Employ machine learning techniques using a Random Forest Classifier to analyze applicant details and predict credit card approval through a
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	Flask-based web application.
Key Features	• Implementation of a machine learning-based credit card approval prediction model. • Real-time prediction of credit card approval or rejection. • Flask web application with HTML/CSS user interface. • Data preprocessing using Pandas and NumPy. • Fast, accurate, and user-friendly prediction system.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources (CPU specifications, number of cores)	Intel Core i5/i7 Processor or equivalent
	Memory (RAM specifications)	8 GB RAM
	Storage (Disk space for data, models, and logs)	256 GB SSD or above
Software	Frameworks	Flask
	Libraries	scikit-learn, pandas, numpy, matplotlib, pickle
	Development Environment (IDE)	Visual Studio Code
Data	Source, size, format	Credit Card Approval Dataset (CSV), Kaggle Dataset, clean_dataset.csv